

Integers, Primes and Roots — Paper C

HARD • NON-CALCULATOR • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to come out exactly.
- The rules for HCF, LCM and signs are not printed on this paper. You are expected to know them.
- You are expected to know the squares up to 15^2 and the cubes up to 10^3 by heart.
- Several questions describe steps you may not have seen before — apply the ideas you already have. If a number will not factorise neatly, ask whether its prime factorisation is telling you something about squares or cubes.
- The mark for each question is shown in square brackets on the right.
- If a question stops you, leave it and come back — do not spend more than five minutes stuck.

SECTION 1 – WHAT THE PRIME FACTORISATION TELLS YOU

1. Consider the number 1176.

- (a)** Write 1176 as a product of its prime factors, in index form. [2]

- (b)** Hence state, with a reason, whether 1176 is a perfect square. Your reason must refer to the indices. [2]

2. Consider the number 1080.

- (a)** Write 1080 as a product of its prime factors, in index form. [1]

(b) Find the smallest positive whole number k such that $1080k$ is a perfect square. [2]

(c) Hence write down the value of $\sqrt{1080k}$. You should not need to work out $1080k$ first. [1]

3. Using your prime factorisation of 1080 again:

(a) find the smallest positive whole number m such that $1080m$ is a perfect cube [2]

(b) write down the cube root of $1080m$ [1]

SECTION 2 – WORKING BACKWARDS

4. Two whole numbers have an HCF of 8 and an LCM of 96.

(a) Write down the product of the two numbers. [1]

(b) One of the numbers is 24. Find the other, and verify your answer by finding the HCF and LCM of your pair. [2]

- (c) There is one other pair of whole numbers with this HCF and this LCM. Write it down. [1]

5. The HCF of 12 and n is 4, and the LCM of 12 and n is 60. Find n , and verify your answer using prime factorisations. [3]

6. Work out:

(a) $(-3)^2 \times (-2)^3$ [1]

(b) $-60 \div -4 \div -5$ [1]

(c) $(-2)^6$ [1]

(d) $-4^2 + (-4)^2$ [1]

SECTION 3 – REASONING IT OUT

7. Three lights on a tower flash at intervals of 6, 10 and 15 seconds. They all flash together at exactly 12:00:00.

(a) After how many seconds do all three next flash together? [1]

(b) How many more times do all three flash together during the next 5 minutes? [1]

(c) A fourth light, flashing every 8 seconds, also flashed at 12:00:00. How long until all four flash together? Give your answer in minutes. [2]

8. This question is about $\sqrt{300}$, and no calculator may be used.

(a) Show that $\sqrt{300}$ lies between 17 and 18. [1]

(b) State which of 17 and 18 it is nearer to, and justify your answer with a calculation. [1]

(c) Estimate $\sqrt{300}$ to one decimal place. You must justify your estimate by squaring. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.